

REQUIREMENT ANALYSIS

Technology Stack

Date	01 JULY 2026
Team ID	
Project Name	Strategic Product Placement Analysis
Maximum Marks	4 Marks

Introduction

The **Strategic Product Placement Analysis** project is developed using Tableau Desktop for data visualization and Tableau Public for dashboard publishing. The project analyzes retail sales data stored in a CSV file to generate interactive dashboards and business insights.

The technology stack has been selected to ensure efficient data visualization, easy deployment, and user-friendly business analytics. GitHub is used for project version control and documentation, while Microsoft Word and PDF are used for preparing project reports.

Technical Architecture

The technical architecture represents the overall workflow of the project from data collection to business decision-making.

Architecture Flow

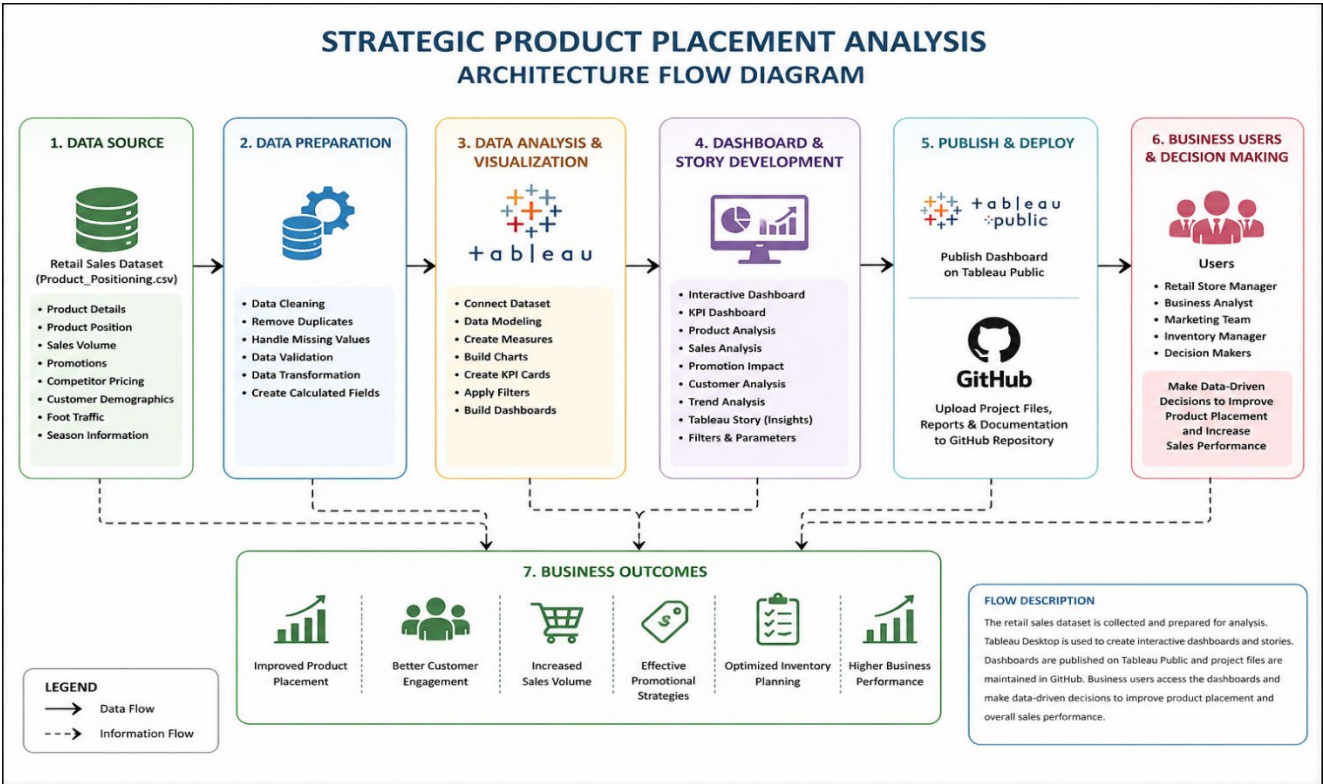


Table-1: Components & Technologies

S. No	Component	Description	Technology Used
1	User Interface	Interactive dashboard used by retail managers and business analysts.	Tableau Dashboard
2	Application Logic	Processes retail sales data and generates visual analytics.	Tableau Desktop
3	Data Preparation	Cleans and prepares the CSV dataset before visualization.	Tableau Desktop (Data Source)
4	Visualization Engine	Creates charts, KPI cards, filters, dashboards, and stories.	Tableau Desktop
5	Dataset	Stores retail sales information for analysis.	CSV File
6	Dashboard Publishing	Publishes dashboards for online access.	Tableau Public
7	Version Control	Maintains project files and documentation.	GitHub
8	Documentation	Prepares reports and supporting documents.	Microsoft Word, PDF
9	Operating System	Platform used for dashboard development.	Windows 10/11
10	Hardware	Computer used for development and analysis.	Personal Computer / Laptop

Table-2: Application Characteristics

S. No	Characteristic	Description	Technology
1	Data Visualization	Interactive retail sales dashboards and charts.	Tableau Desktop
2	Dashboard Publishing	Online dashboard sharing and accessibility.	Tableau Public
3	Data Storage	Retail sales dataset storage.	CSV File
4	Documentation	Project reports and documentation.	Microsoft Word, PDF
5	Version Control	Source file management and collaboration.	GitHub
6	Performance	Fast dashboard rendering with interactive filters.	Tableau Desktop

S. No	Characteristic	Description	Technology
7	Availability	Dashboard accessible online through Tableau Public.	Tableau Public
8	Scalability	Supports larger retail datasets and additional dashboards.	Tableau Platform
9	Reliability	Accurate business insights from validated retail data.	Tableau Desktop
10	Maintainability	Easy updates for dashboards and documentation.	GitHub & Tableau

Software Requirements

Software	Purpose
Tableau Desktop	Dashboard development
Tableau Public	Dashboard publishing
GitHub	Project repository
Microsoft Word	Documentation
PDF Reader	Viewing reports

Hardware Requirements

Hardware	Specification
Processor	Intel Core i3 or above
RAM	8 GB or higher
Storage	10 GB Free Space
Operating System	Windows 10/11
Internet	Required for Tableau Public & GitHub

Advantages of the Selected Technology Stack

- Interactive and user-friendly dashboards.
- Easy visualization of retail sales data.
- Online dashboard publishing through Tableau Public.

- Simple project management using GitHub.
 - Supports future scalability.
 - Requires minimal coding knowledge.
 - Suitable for business intelligence applications.
-

References

- [Tableau Desktop Documentation](#)
 - [Tableau Public Documentation](#)
 - [GitHub Documentation](#)
 - [Microsoft Office Documentation](#)
-